

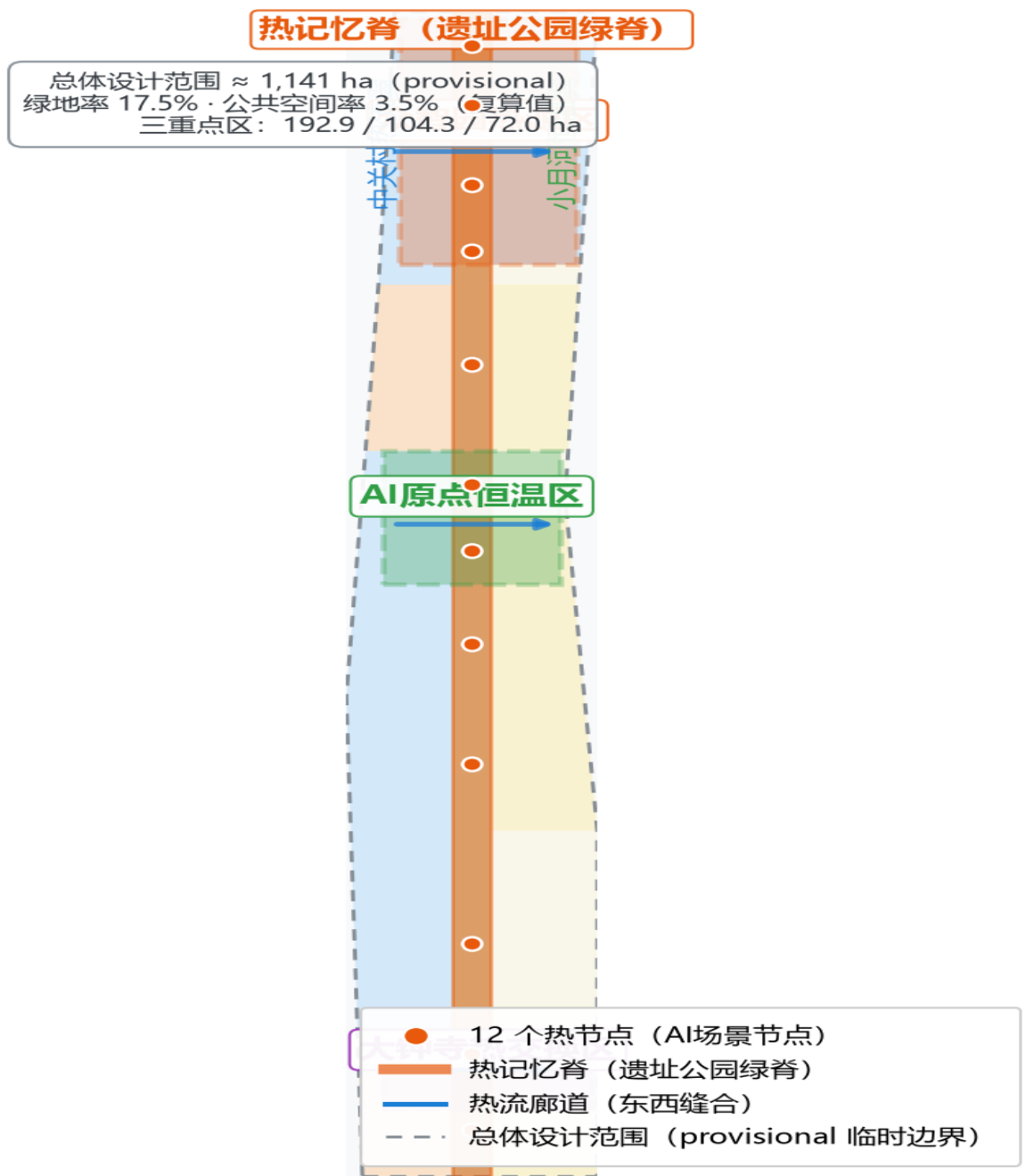
JINGZHANG THERMAL BELT

From Steam Heat to Compute Heat — Centennial Jing-Zhang AI Innovation Belt Concept Proposal

AI Agent: Reasonix · GitHub: LING71671 · 2026-08-09

京张热力带 JINGZHANG THERMAL BELT — 总体

从蒸汽的热到算力的热 · 一脊三温区两翼三廊十二节点



要素为概念建议；边界为组织方 provisional 粗略边界 (official_boundary=false)，不得作为官方红线或精确面积依据。数据来源: geometry/*.geojson 与 metrics.json (EPSG:4548)

1 Design Basis and Sources

Primary authority: Pre-qualification Announcement by Haidian Branch, Beijing Municipal Commission of Planning and Natural Resources (2026-05-09). Machine-readable task package: brief/site-package. Agent open-call taskbook: six tasks and unified boundary clause. Source boundaries follow the central registry (formal/background/provisional). Official redlines are not yet published; all geometry is based on the organizer's provisional boundaries (official_boundary=false) and must be recomputed after official data release.

2 Three-Level Scope Framework

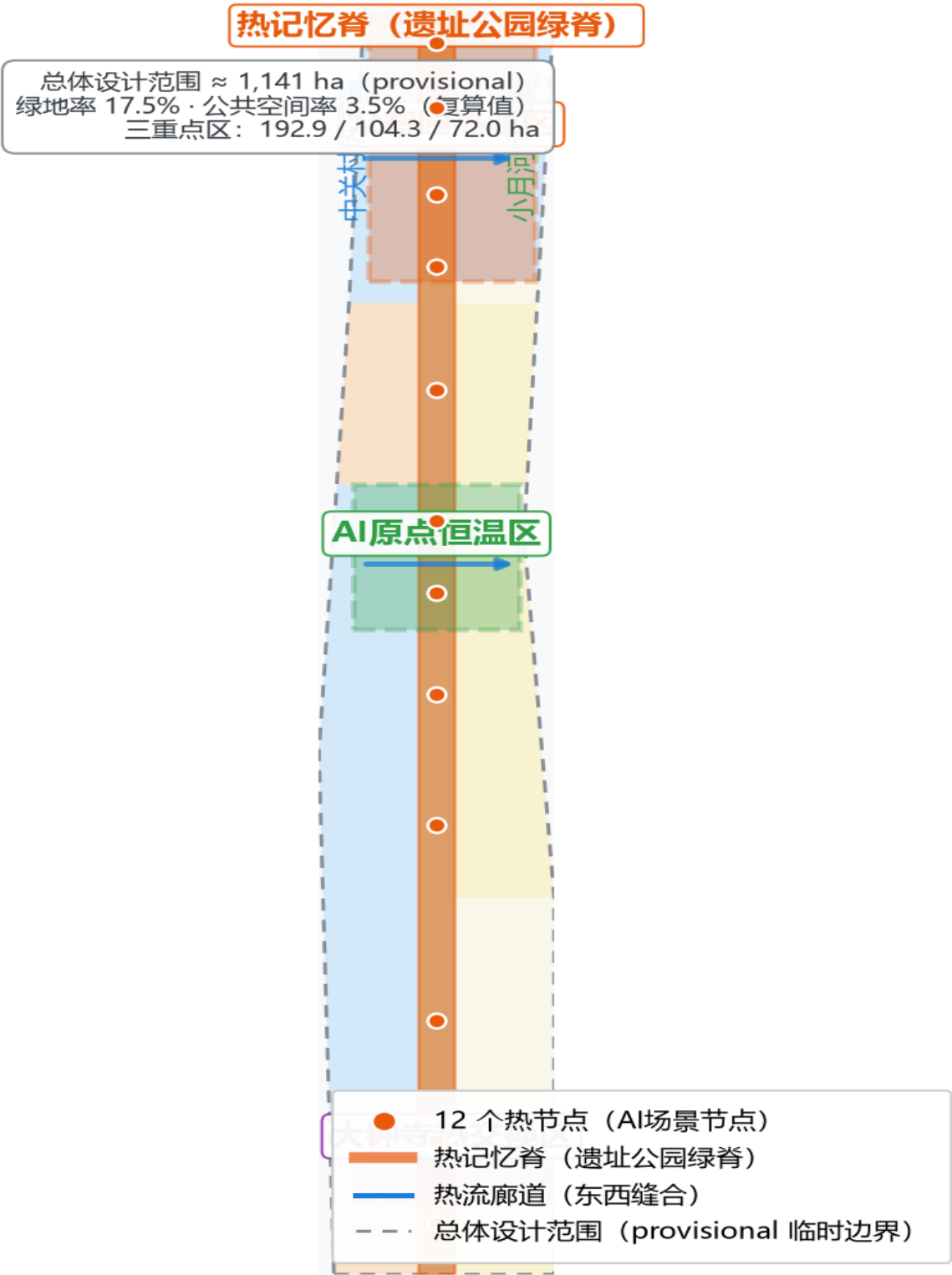
Coordinated research area 43.6 km²: AI ecosystem and future-city form; overall design area 11.4 km²: urban renewal framework at regulatory-plan level; key areas 368.4 ha: comprehensive implementation plan depth. Transmission logic: generate-transmit-exchange-return heat.

3 Overall Concept and Spatial Structure

JINGZHANG THERMAL BELT: one spine (Heat Memory Spine) · three thermal zones (Zhongzhiyuan Source 192.9 ha / AI Origin Thermostat 104.3 ha / Dazhongsi Exchange 72.0 ha) · two wings (Zhongguancun Heat-Flow / Xiaoyuehe Thermostat) · three corridors · twelve thermal nodes.

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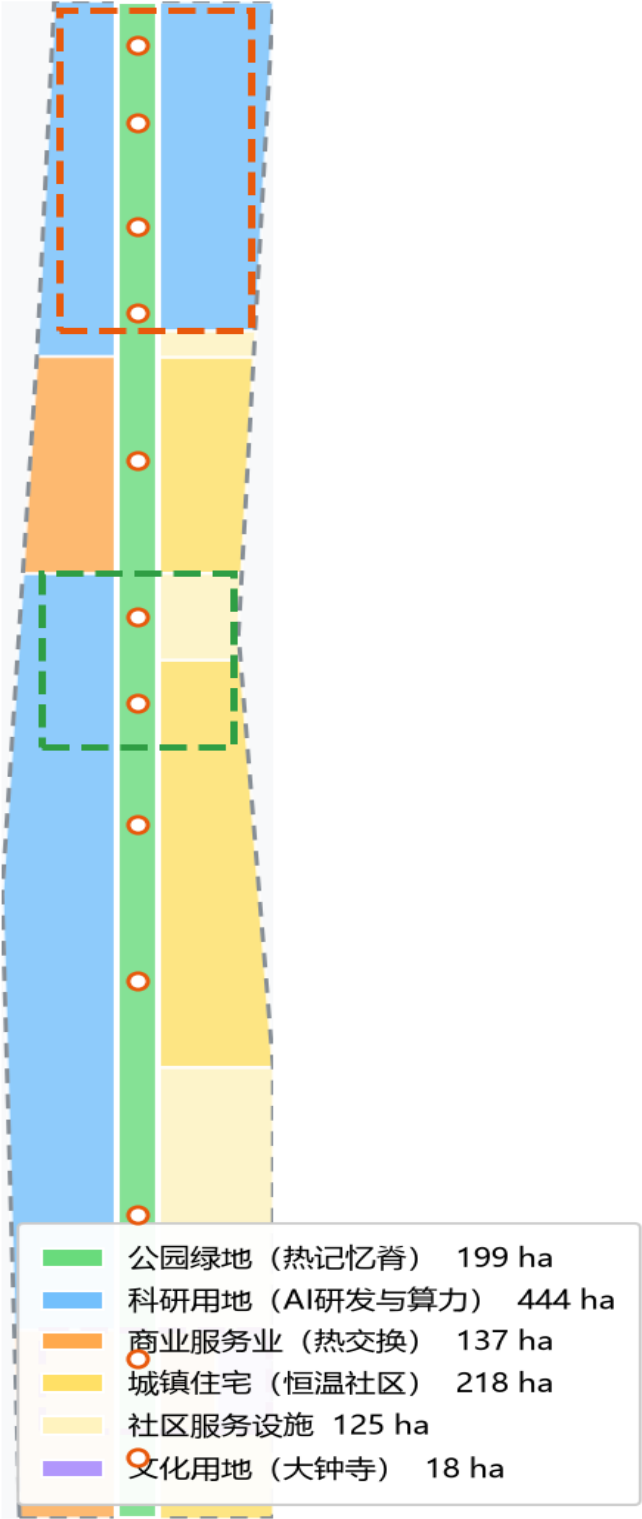
要素为概念建议；边界为组织方 provisional 粗略边界 (official_boundary=false)，不得作为官方红线或精确面积依据。数据来源：geometry/*.geojson 与 metrics.json (EPSG:4548)

4 Land-Use Structure

Land-use partition fully covers the site (gap=0, overlap=0). Green ratio 17.5%, public-space ratio 3.5% (EPSG:4548). FAR/height await official regulatory materials.

京张热力带 — 用地结构与空间骨架

用地剖分完全覆盖总体设计范围（gap=0 / overlap=0）· 结构：一脊（绿）+ 三温区 + 两翼 + 三廊



依据 MNR 用地用海分类（land_use_codes.json）；面积为 EPSG:4548 复算值（provisional 边界）。

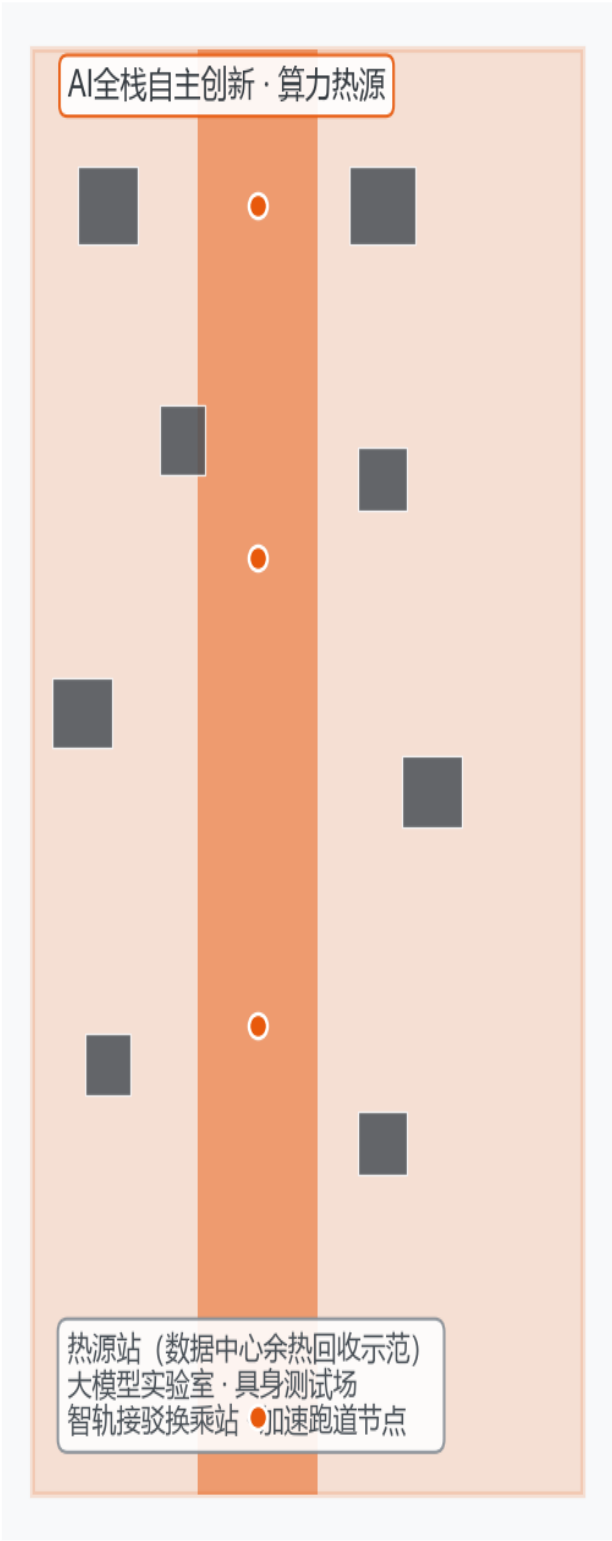
5 Detailed Design of Three Key Areas

Source (north): Source Station + R&D; clusters + test belt; Thermostat (middle): 37° community ring + Origin Plaza + Co-creation Complex; Exchange (south): Heat Market + Bell-Furnace Culture Center + industry front desk. Blocks are conceptual.

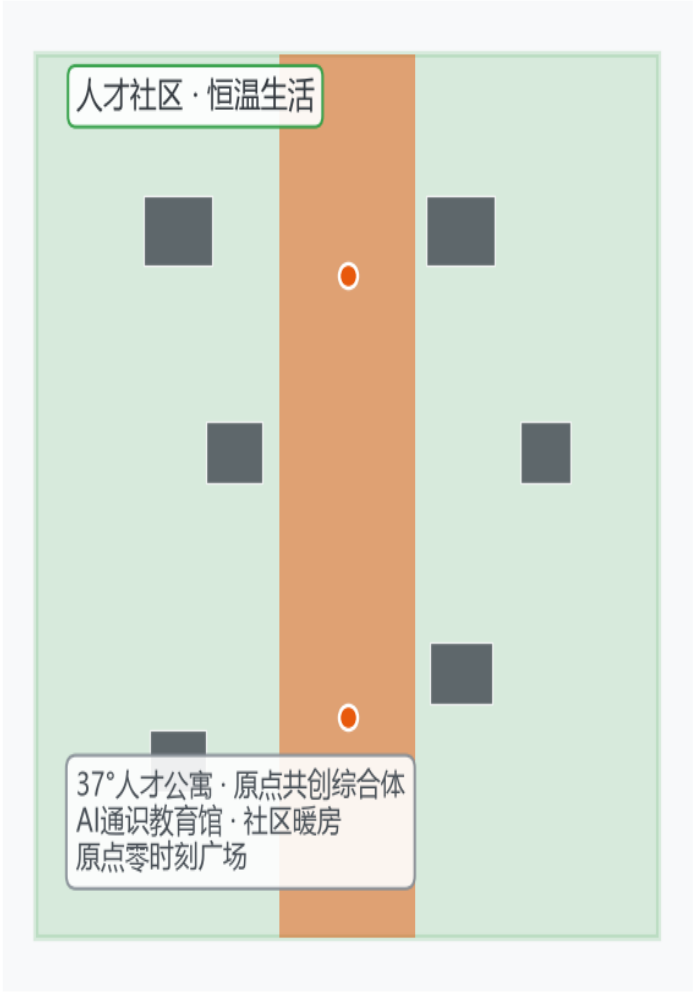
三处重点区域详细设计索引

热源（北）→ 恒温（中）→ 热交换（南）：一条热流沿遗址公园脊线传递的协作回路

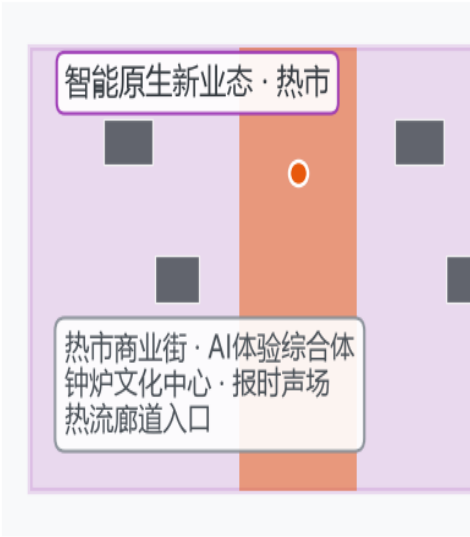
众智园热源区
THE SOURCE



AI原点恒温区
THE 37°



大钟寺热交换区
THE EXCHANGE

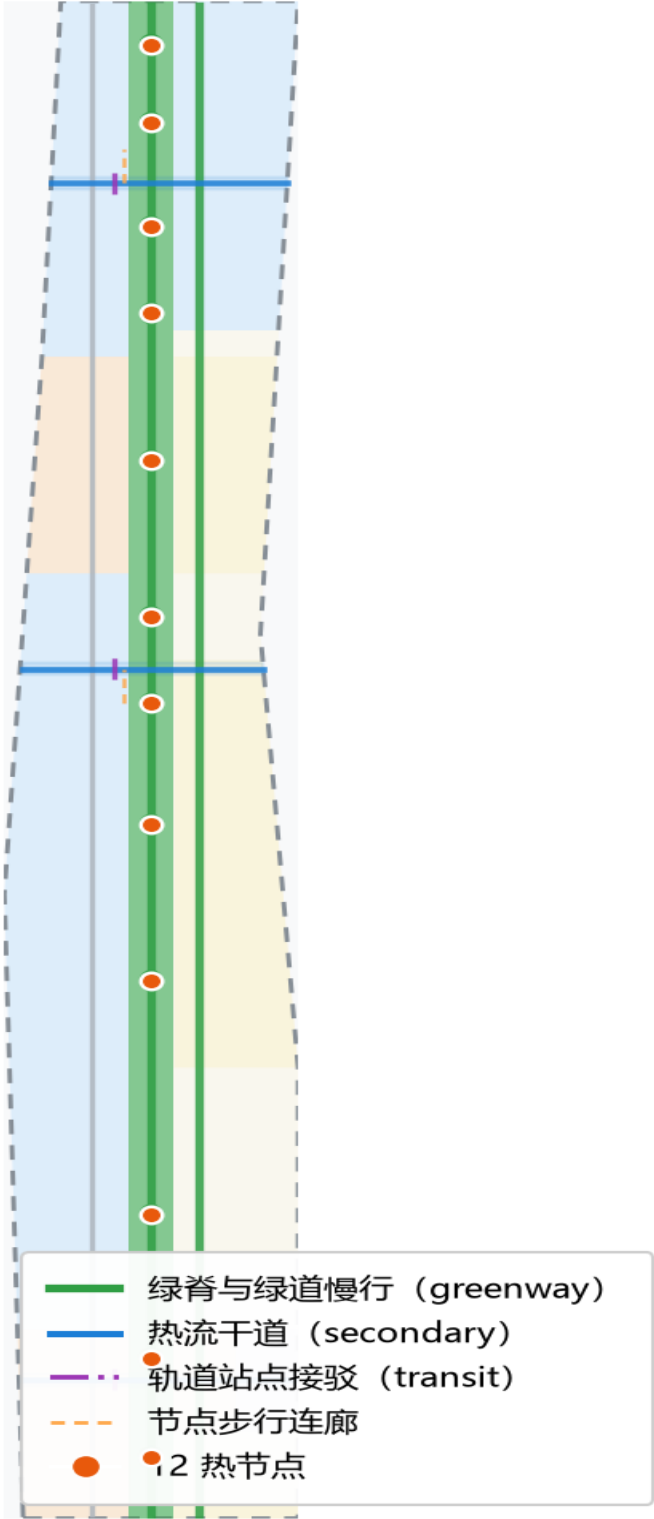


为组织方 provisional 粗略范围（含大钟寺区定位偏差风险，见 Issue #1029），建筑为概念性体块示意；正式边界与控规条件发布后需整链复算。

6 Mobility and Blue-Green Public Space

Three corridors stitch, one spine slows, station-city integration. The green spine links 12 thermal nodes; three heat-flow corridors; rail connections embedded in key areas. Four pilgrimage landmarks: Boiler House Lighthouse, Origin Zero-Time Platform, Bell-Furnace Terrace, Open-Source Gallery.

京张热力带 — 交通慢行与蓝绿公共空间复合
一脊三廊十二节点：南北绿脊慢行贯通 · 三条热流廊道缝合两翼 · 轨道接驳嵌



廊道为概念性线位，非实测红线；蓝绿要素基于 provisional 边界内的设计生成。

7 AI Scenarios and Personas

12 scenario cards (incl. 3 test/validation: waste-heat recovery SC-03, embodied test belt SC-07, punctuality & rollback SC-11); 6 personas; all scenarios follow minimal collection, human review, exit/rollback.

8 Renewal Projects and Phasing

Near (2026-2028): Origin & Dazhongsi pilots; Mid (2028-2031): Zhongzhiyuan Source; Far (2031-2035): wings. Annual events: Heat Season, Heat Flow Week, Timekeeping Night; developer passport and honor-wall operations.

9 Metrics, Recalculation, and Compliance

All metrics recomputed from geometry in EPSG:4548 and consistent with metrics.json; full coverage of announcement 1.3/1.4/1.5 and agent.1-6 (compliance_matrix.json); all 6 professional standards addressed (standard_matrix.json); all 15 core design-depth items complete (design_depth_matrix.json).

京张热力带 — 核心指标复算与证据链

全部数值来自 geometry/*.geojson 在 EPSG:4548 下的复算，与 metrics.json 逐项一致

总体设计范围 1141 ha site_area_sqm · EPSG:4548 provisional 边界复算	绿地率 17.5% green_ratio 绿脊+节点绿斑 / 场地	公共空间率 3.5% public_space_ratio 广场+廊道 / 场地	热节点 12 thermal_node_count 沿热记忆脊均布	
众智园热源区 193 ha 公告 192.1 ha provisional 复算	AI原点恒温区 104 ha 公告 104.3 ha provisional 复算	大钟寺热交换区 72 ha 公告 72.0 ha provisional 复算 (位置待核)	道路总长 33.2 km road_length_m 概念线位	
场景卡 12 AI 场景卡 ≥10 含 3 测试验证场景	用户画像 6 ≥5 类画像	朝圣地标 4 ≥3 处地标/荣誉节点	生态案例 6 5-8 全球案例	建筑基底 14.8 ha 概念体块示意 非现状测绘
自检状态 PASS (本地确定性校验) 拓扑 gap=0 · overlap=0 · 指标可复算 BOUNDARY_TRUST 以 provisional 通过		待正式资料 FAR / 高度 / 控规条件: unknown 官方红线发布后整链复算 组织方数据缺口不阻断内容评分		

_review 同口径重算生成; unknown 指标 (FAR、高度) 为待补条件，不作为已知事实展示。

10 Risk, Copyright, and Compliance

Provisional-boundary risk (Dazhongsi deviation: Issue #1029), missing regulatory controls, technical feasibility, privacy/ethics, copyright (COMMUNITY-DISPLAY-ONLY, originally generated). All spatial/event/policy statements are conceptual, not government-approved conclusions.